

ARTICLE OPEN ACCESS

Transporter-Drive Interstitial Tissue Exposure and Pharmacodynamic Response of Meropenem in Sepsis: A Mechanistic PBPK Modeling Study

Laura Ben Olivo¹  | Jessica Luisa Silva de Lemos² | Vinicius Jardim Rodrigues² | Bibiana Verlindo de Araújo^{1,2} 

¹Pharmaceutical Sciences Graduate Program, Federal University of Rio Grande Do Sul, Porto Alegre, Brazil | ²Biological Sciences Graduate Program: Pharmacology and Therapeutics, Federal University of Rio Grande Do Sul, Porto Alegre, Brazil

Correspondence: Bibiana Verlindo de Araújo (bibiana.araujo@ufrgs.br)

Received: 7 May 2026 | **Revised:** 23 July 2026 | **Accepted:** 21 August 2026

Keywords: PBPK/PD modeling | sepsis | tissue exposure

ABSTRACT

A whole-body mechanistic PBPK model for meropenem (MPN) was developed in PK-Sim and validated using published plasma and tissue concentration–time data in healthy volunteers and critically ill patients. Renal elimination was implemented as glomerular filtration plus active tubular secretion (basolateral uptake via OAT3 with apical efflux), and non-renal clearance via DHP-mediated hydrolysis. The model was scaled to sepsis or septic shock by incorporating disease-specific physiological changes and optimizing OAT3 activity and tissue permeability to reproduce observed variability. Simulations in a virtual septic population assessed unbound interstitial concentrations in clinically relevant tissues under ILAS-recommended typical and maximum dosing MPN regimens. Antibacterial effect against *Escherichia coli* and *Klebsiella pneumoniae* was evaluated using a published PK/PD model driven by simulated unbound tissue concentrations. The PBPK model reproduced observed plasma profiles in healthy volunteers and captured plasma and subcutaneous interstitial exposure in sepsis. Simulations showed clear dissociation between plasma and interstitial exposure and marked tissue-specific heterogeneity. Predicted clearance increased in sepsis (augmented renal clearance) and decreased in septic shock (impaired renal function and secretion). Although maximum dosing increased plasma and tissue exposure, PD simulations indicated effects were already near the plateau with standard dosing, yielding minimal additional antibacterial benefit from routine dose escalation. Transporter-informed PBPK/PD modeling explains dynamic, severity-dependent changes in meropenem clearance and tissue exposure in sepsis, highlights limitations of plasma-only assessment, and supports individualized, mechanism-informed optimization rather than universal dose scaling.

1 | Introduction

Sepsis remains a leading cause of morbidity and mortality in critically ill patients, and prompt administration of effective antimicrobial therapy is a cornerstone of its management. Meropenem (MPN), a broad-spectrum β -lactam antibiotic, is frequently used as first-line therapy in severe infections due to its favorable safety profile and extensive antimicrobial coverage. Despite the implementation of aggressive dosing strategies and therapeutic drug monitoring, clinical failure continues to be observed in

patients with sepsis, even when plasma pharmacokinetic targets are apparently achieved. This discrepancy underscores a fundamental limitation of plasma-centered pharmacokinetic assessments in complex disease states such as sepsis [1–3].

From a pharmacodynamic (PD) perspective, antimicrobial efficacy is driven by the unbound drug concentration at the site of infection, most accurately represented by the interstitial fluid (ISF) of target tissues. In critically ill patients, sepsis-induced alterations in regional perfusion, endothelial permeability, and

This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial](https://creativecommons.org/licenses/by-nc/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited and is not used for commercial purposes.

© 2026 The Author(s). CPT: Pharmacometrics & Systems Pharmacology published by Wiley Periodicals LLC on behalf of American Society for Clinical Pharmacology and Therapeutics.

Study Highlights

- What is the current knowledge on the topic?
 - Meropenem pharmacokinetics is highly variable in critically ill patients due to sepsis-related physiological changes such as augmented renal clearance and acute kidney injury. Plasma concentrations are commonly used to guide therapy but may not reflect drug exposure at infection sites. In addition, the contribution of transporters to carbapenem disposition is rarely incorporated into clinical pharmacokinetic models.
- What question did this study address?
 - This study evaluated how transporter-mediated processes and sepsis severity influence meropenem interstitial tissue exposure and antibacterial response, and whether empirical dose escalation improves pharmacodynamic outcomes.
- What does this study add to our knowledge?
 - A transporter-informed PBPK/PD model predicts tissue-resolved meropenem exposure across sepsis severity and demonstrates a dissociation between plasma and interstitial concentrations. Simulations indicate that dose escalation increases exposure but yields minimal additional antibacterial benefit when the pharmacodynamic effect approaches saturation.
- How might this change drug discovery, development, and/or therapeutics?
 - Mechanistic PBPK models integrating transporters and disease physiology may improve prediction of tissue antibiotic exposure and support individualized dosing strategies rather than routine empirical dose escalation in sepsis.

microvascular function may decouple plasma exposure from tissue exposure, rendering plasma concentrations an unreliable surrogate for drug availability at the effect site [1]. These pathophysiological changes evolve dynamically during sepsis and differ across organs, highlighting the need for tissue-specific evaluation of antimicrobial exposure.

Clinical studies employing tissue-based sampling techniques have demonstrated substantial heterogeneity in MPN distribution beyond the central compartment. Investigations of intrapulmonary exposure in critically ill patients revealed pronounced interindividual variability in epithelial lining fluid and lung tissue concentrations, even under optimized dosing strategies such as continuous infusion [4, 5]. Moreover, studies directly comparing plasma and interstitial MPN concentrations indicate that equivalence between compartments cannot be assumed, particularly in patients receiving continuous renal replacement therapy or presenting with severe physiological disturbances [6]. This concept is further supported by more recent reviews highlighting the limitations of plasma concentrations as surrogates for target-site exposure in critical illness [1].

In addition to distributional changes, sepsis profoundly affects MPN clearance. Augmented renal clearance (ARC) has been

increasingly recognized in septic and critically ill populations and represents a major driver of subtherapeutic β -lactam exposure, even when conventional or intensified dosing regimens are applied [7]. While increases in glomerular filtration contribute to this phenomenon, they do not fully explain the marked variability observed in MPN exposure across tissues, suggesting the involvement of additional mechanistic pathways.

Among these pathways, membrane transporters play a pivotal role in determining both drug elimination and tissue exposure. MPN has been identified as a substrate of key renal organic anion transporters, including OAT1 and OAT3, implicating active tubular secretion as an important contributor to its systemic clearance [8]. Transporter-mediated processes may also influence intracellular and interstitial drug exposure within organs involved in elimination and inflammation-driven redistribution, such as the kidneys, liver, and lungs.

Importantly, inflammatory states characteristic of sepsis may modulate transporter expression and activity. Experimental models of infection and systemic inflammation have demonstrated dysregulation of renal uptake and efflux transporters, providing a plausible mechanistic basis for disease-dependent alterations in MPN disposition that extend beyond traditional perfusion-limited assumptions [9]. These observations support the hypothesis that transporter-mediated mechanisms contribute meaningfully to variability in MPN interstitial tissue exposure during sepsis, yet such mechanisms are rarely incorporated explicitly into clinical pharmacokinetic analyses.

Physiologically based pharmacokinetic (PBPK) modeling offers a robust mechanistic framework to integrate organ-specific physiology, disease-related alterations, and transporter-mediated drug disposition. By explicitly representing tissue compartments, regional blood flows, and active transport processes, PBPK models enable quantitative prediction of drug exposure in plasma and multiple organs under both physiological and pathological conditions. This approach is particularly well suited to sepsis, where simultaneous changes in perfusion, permeability, renal function, and transporter activity occur across tissues [10].

In this study, we apply a mechanistic PBPK modeling approach incorporating transporter-mediated processes to characterize MPN disposition and PD response across clinically relevant organs, including muscle, subcutaneous tissue, lung, liver, and kidneys in sepsis condition. By integrating sepsis-associated physiological alterations with active transport mechanisms and a concentration–effect PD model, this work aims to elucidate the determinants of tissue-specific MPN exposure and to assess how PK variability translates into antibacterial effect. Together, this framework advances a mechanistic understanding of β -lactam PK and PD in critically ill patients.

2 | Methods

2.1 | Clinical Data Used for Model Development and Validation

Clinical data used for the development and validation of the PBPK models were obtained from previously published

clinical studies reporting MPN pharmacokinetics in healthy volunteers [11–16] and patients with sepsis [17] and septic shock [18]. Only peer-reviewed studies providing quantitative concentration–time data in plasma and/or tissues were considered. Data sources were selected to ensure coverage of both systemic exposure and target-site distribution under clinically relevant conditions. Data points were extracted using WebPlotDigitalizer [19].

2.2 | PBPK Model for Healthy Volunteers

Default physiological parameters implemented in PK-Sim were used to represent healthy adult volunteers. MPN-specific physicochemical properties employed for compound definition were extracted from Chemicalize software. Renal and non-renal elimination of meropenem were explicitly implemented in the PBPK model. MPN renal elimination was modeled mechanistically as the combined contribution of glomerular filtration (GFR) and active tubular secretion, the latter represented by basolateral uptake via renal organic anion transporters (OAT3) followed by apical transport into the tubular lumen, while non-renal clearance was accounted for by dehydropeptidase-1 (DHP)-mediated hydrolysis (Figure 1). Physicochemical and elimination-related input parameters for meropenem are summarized in Table 1.

Model performance was evaluated using the average fold error (AFE) (Equation 1) to compare observed and predicted concentrations, with AFE quantifying the overall bias of model predictions. In addition, fold error (FE) (Equation 2) was calculated for key pharmacokinetic parameters, including clearance (CL), volume of distribution (V_d), maximum concentration (C_{max}), and area under the concentration–time curve (AUC). Model predictions were considered acceptable when FE and AFE values fell within a 0.5- to 2.0-fold range.

$$AFE = 10^{\frac{1}{n} \sum \log_{10} \left(\frac{C_{pred}}{C_{obs}} \right)} \quad (1)$$

$$FE = \frac{PK \text{ parameter}_{Obs}}{PK \text{ parameter}_{Pred}} \quad (2)$$

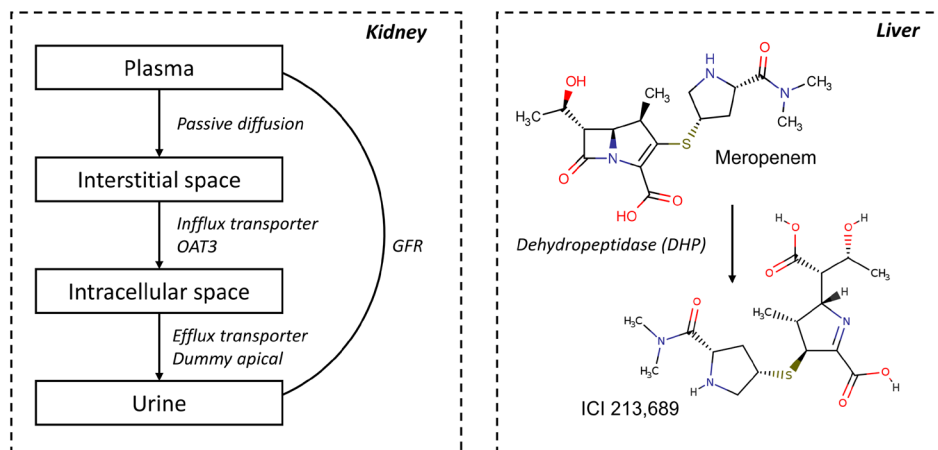


FIGURE 1 | Mechanistic representation of meropenem elimination pathways implemented in the PBPK model.

2.3 | Scaling PBPK Model for Patients With Sepsis and Septic Shock

The PBPK model developed for healthy adult volunteers was scaled to represent patients with sepsis and septic shock by incorporating disease-specific physiological and mechanistic alterations. Sepsis and septic shock were treated as distinct clinical states, reflecting the dynamic and progressive nature of the condition. Physiological parameters were adjusted in accordance with associated changes reported by Radke et al. [21].

In hyperdynamic sepsis, GFR fraction was increased to reflect augmented renal clearance commonly observed in critically ill patients. In contrast, septic shock was represented by reductions in GFR, consistent with impaired renal function and the development of acute kidney injury. Active renal elimination of meropenem under disease conditions was modeled mechanistically by optimizing the catalytic turnover rate (kcat) of basolateral uptake via renal OAT3, reflecting inflammation-associated alterations in transporter activity during severe infection. Apical transport into the tubular lumen was retained as a non-limiting step. Non-renal clearance mediated by DHP was maintained across disease states and assumed to be unaffected by sepsis severity.

Tissue distribution was evaluated in sepsis by combining effects of altered renal function, transporter-mediated processes, and endothelial permeability (P_{end}). Similarly, to the optimization of the OAT3 kcat, P_{end} of subcutaneous tissue was optimized using the Levenberg–Marquardt algorithm [22] to improve prediction of unbound interstitial concentrations. Both optimizations were performed against concentration–time data of plasma (kcat) and subcutaneous tissue (P_{end}). Model evaluation procedures and acceptance criteria were identical to those applied in the healthy volunteer PBPK model.

To characterize the individual contributions of glomerular filtration and active tubular secretion to MPN renal elimination, a local sensitivity analysis (SA) was performed by varying GFR fraction and OAT3 kcat independently over a range of $\pm 50\%$ relative to their optimized values, with all other parameters held constant.

TABLE 1 | Input parameters describing physicochemical characteristics and mechanistic elimination pathways of meropenem in the PBPK model.

Parameter	Healthy volunteers		Septic patients		Shock Septic patients		References
	Value		Value		Value		
Molecular weight (g/mol)	383.46		383.46		383.46		Chemicalize
Log P	−2.96		−2.96		−2.96		Optimized by PK-Sim
pKa	3.73 (acid), 8.39 (base)		3.73 (acid), 8.39 (base)		3.73 (acid), 8.39 (base)		Chemicalize
Solubility at pH 7 (mg/L)	5.63		5.63		5.63		Chemicalize
Fraction unbound	0.98		0.98		0.98		[20]
Partition coefficients	Rodgers and Rowland		Rodgers and Rowland		Rodgers and Rowland		[20]
Cellular permeabilities	Charge dependent Schmitt		Charge dependent Schmitt		Charge dependent Schmitt		[20]
P plasma ↔ interstitial (cm/min)	100 ^a		2.4 × 10 ^{−4}		2.4 × 10 ^{−4}		Optimized by PK-Sim
Transporter and excretion							
GFR fraction	1.0		3.5		0.8		Optimized by PK-Sim
<i>OAT3</i>							
Initial Conc (μM)	0.09		0.09		0.09		[20]
Km (μM)	847.0		847.0		847.0		[20]
Kcat 1/min	3548.9		525.98		300.85		Optimized by PK-Sim
<i>APICAL</i>							
Initial Conc (μM)	1.00		1.00		1.00		[20]
Km (μM)	770.0		770.0		770.0		[20]
Kcat 1/min	128.56		128.56		128.56		[20]
<i>DHP</i>							
Km (mM)	2.0		2.0		2.0		[20]
<i>V</i> _{max} (μmol/L/min)	8.4		8.4		8.4		Optimized by PK-Sim

^aPK-Sim default value (Open Systems Pharmacology Suite Manual, <https://docs.open-systems-pharmacology.org>).

Sensitivity was quantified as the normalized change in AUC_{inf} and C_{max} in plasma and interstitial compartments of kidney, liver, lung, muscle, and skin. To evaluate the impact of the unbound P_{end} assumption on model predictions, an analogous SA was performed by varying the endothelial permeability of each simulated tissue independently by $\pm 50\%$ relative to the optimized base value, in both sepsis and septic shock models, comprising nine simulation runs per tissue. Sensitivity was assessed in terms of AUC_{inf} and C_{max} across plasma and all interstitial compartments.

2.4 | Prediction of Interstitial Tissue Exposure and Pharmacodynamic Evaluation

Simulations were performed to predict unbound concentrations for the kidney, lung, liver, and subcutaneous tissue, reflecting their clinical relevance as common sites of infection in both sepsis conditions. Organ-specific physiological alterations associated with sepsis were incorporated into the PBPK model to capture pathophysiological changes affecting drug distribution, with particular emphasis on modifications in capillary permeability and tissue accessibility. P_{end} was considered the same for all simulated tissues.

A virtual population of 100 adult subjects with sepsis or septic shock was generated in PK-Sim, comprising a 1:1 male-to-female ratio and an age range of 50–70 years, with body weights assigned according to default PK-Sim adult distributions for this age range. Interindividual variability was incorporated using lognormal distributions applied to key parameters governing elimination, transporter-mediated processes, and plasma protein binding, assuming a geometric standard deviation of 2.0 to capture clinically relevant variability across the simulated population. The dosing regimen followed the ILAS [23] protocol (Table 2) for general and maximum dosing in sepsis. Unbound interstitial concentration–time profiles were predicted for each tissue, enabling a specific assessment of meropenem distribution under both septic conditions.

Antibacterial effect was evaluated using a previously published PK/PD model of MPN against *Escherichia coli* and *Klebsiella pneumoniae*, encompassing strains with minimum inhibitory concentrations (MICs) ranging from 0.03 to 128 mg/L [24]. The model describes bacterial dynamics using two interconnected subpopulations: drug-susceptible bacteria (S) and dormant bacteria (D). Both bacterial species were explicitly simulated, and identical PD parameters were applied, as no strain-specific differences were reported in the original model.

TABLE 2 | Dosing regimens of meropenem simulated in the final PBPK model.

Protocol	Loading dose (g)	Maintenance dose (g)	Time of infusion (h) [Frequency]
Typical sepsis dose	1.5	1	3 [q8h]
Highest sepsis dose	2	2	3 [q8h]

Bacterial growth and transition between subpopulations were governed by a system of ordinary differential equations, which are fully described in the (Equations S1–S3). Drug-induced bacterial killing was characterized using a sigmoidal E_{max} model (Equation 3), with simulated unbound tissue concentrations serving as the PK input.

$$Effect = \frac{E_{max} \times C^\gamma}{EC_{50}^\gamma + C^\gamma} \quad (3)$$

Effect represents the pharmacodynamic response, E_{max} the maximum achievable antibacterial effect, C the simulated drug concentration, EC_{50} the concentration producing 50% of the maximum effect, and γ the Hill coefficient describing the steepness of the concentration–effect relationship. PD parameters values were fixed according to previously published estimates (Table S1).

Treatment was considered successful when a reduction of at least 2 log units in total bacterial load was achieved over the simulated treatment.

3 | Results

3.1 | PBPK Model for Healthy Volunteers

The PBPK model for MPN adequately described plasma concentration–time profiles in healthy adult volunteers across multiple independent clinical studies (Table S2). The goodness of fit plot from simulated plasma concentrations are shown in Figure S1. The model closely matched observed data, capturing both distribution and elimination phases without systematic bias.

Across studies, the AFE for plasma concentrations ranged from 1.18 to 2.05. For maximum concentration (C_{max}), fold error (FE) values ranged from 0.66 to 1.29, while FE values for area under the concentration–time curve (AUC) ranged from 0.57 to 0.93. The majority of predicted pharmacokinetic parameters fell within the predefined acceptance criteria of 0.5 to 2.0-fold relative to observed values, demonstrating robust predictive performance of the model (Table 3).

The mechanistic implementation of renal elimination via glomerular filtration and active tubular secretion, together with non-renal clearance through DHP-mediated hydrolysis, yielded physiologically plausible estimates of total clearance (~ 0.17 L/h/kg) across all datasets. The PBPK model consistently preserved mass balance, with approximately 70% of the administered dose eliminated unchanged in urine. As a whole, these findings demonstrate that the healthy volunteer PBPK model reliably reproduces systemic meropenem pharmacokinetics and provides a robust baseline for disease-specific scaling to sepsis and septic shock.

3.2 | Scaling PBPK Model for Patients With Sepsis and Septic Shock

Following disease-specific scaling, the PBPK model adequately described MPN pharmacokinetics in patients with sepsis across different dosing strategies and tissues. The goodness of fit plot from simulated plasma and unbound interstitial concentrations

is shown in Figure S2. Model predictions captured both plasma and interstitial concentration–time profiles under short infusion and continuous infusion regimens.

For plasma concentrations, AFE values ranged from 1.12 to 2.13, indicating acceptable overall predictive bias across studies. FE values for plasma $C_{\max}$ ranged from 0.91 to 1.73, while FE values for plasma AUC ranged from 0.97 to 1.64, remaining within the predefined acceptance criteria of 0.5- to 2.0-fold. Prediction of interstitial exposure in subcutaneous tissue demonstrated higher variability compared with plasma. For subcutaneous concentrations, AFE values ranged from 1.75 to 2.13, with FE values for $C_{\max}$ between 0.66 and 0.67, and FE values for AUC between 0.52 and 0.74. All results are compiled in Table 4. The higher variability observed for subcutaneous interstitial predictions likely reflects the combined contribution of microdialysis probe recovery uncertainty, sparse calibration data, and the marked pathophysiological heterogeneity in peripheral tissue perfusion and fluid dynamics characteristic of sepsis.

Across all sepsis simulations, the PBPK model preserved mass balance and maintained physiologically plausible elimination pathways, with renal excretion remaining the dominant route of MPN elimination. Incorporation of disease-specific changes in GFR, together with optimization of OAT3 k_{cat} and subcutaneous P_{end} , was required to reproduce the observed variability in systemic and interstitial exposure without modification of organ blood flow parameters.

SA demonstrated distinct and tissue-specific effects for all evaluated parameters (Figures S3 and S4). For renal elimination

parameters, GFR fraction was the dominant determinant of AUC_{inf} and $C_{\max}$ across all compartments, while OAT3 k_{cat} showed a smaller but consistent contribution most pronounced in the kidney and near-negligible in plasma, supporting the mechanistic separability of these two elimination pathways. For P_{end} , tissue exposure was sensitive primarily to the permeability of the corresponding tissue, with negligible cross-tissue effects. AUC_{inf} remained largely insensitive to P_{end} variation across all tissues and disease states, while $C_{\max}$ showed tissue-specific sensitivity with divergent directionality between disease states, positive in sepsis, where plasma-to-tissue influx dominates at peak, and negative in septic shock, where prolonged plasma exposure favors efflux. The kidney was a mechanistic exception, exhibiting positive AUC_{inf} sensitivity in sepsis, attributable to net interstitial accumulation under saturable OAT3-mediated secretion. Sensitivity magnitudes were consistently lower in septic shock than in sepsis across all analyses.

Overall, these results demonstrate that the sepsis-scaled PBPK model extends the healthy volunteer framework to critically ill patients, providing a mechanistically consistent description of MPN plasma and tissue exposure across clinically relevant dosing regimens.

3.3 | Prediction of Interstitial Tissue Exposure and Pharmacodynamic Evaluation

PBPK simulations revealed marked differences in MPN systemic pharmacokinetics and tissue exposure across disease severity and dosing strategies (Figure 2). In patients with sepsis, typical dosing resulted in lower plasma exposure, whereas administration of the

TABLE 3 | Average fold error and fold error of the model across validation studies for healthy volunteers.

Study	AFE	$C_{\max}$ (mg/L)			AUC (mg.h/L)		
		Pred	Obs	FE	Pred	Obs	FE
Bax et al. [11]	1.27	56.68	49.42	0.87	82.93	68.28	0.82
Christensson et al. [12]	1.70	28.24	32.00	1.13	40.93	37.25	0.91
Krueger et al. [13]	1.52	28.24	31.17	1.10	40.93	38.13	0.93
Leroy et al. [15]	2.05	28.24	18.57	0.66	40.93	23.27	0.57
Ljungberg et al. [14]	1.46	28.24	36.34	1.29	40.93	37.52	0.92
Nilsson-Ehle et al. [16]	1.18	56.68	53.54	0.94	82.93	69.92	0.84

TABLE 4 | Average fold error and fold error of the model across validation studies for patients with sepsis and septic shock.

Study	Tissue	Protocol	AFE	$C_{\max}$ (mg/L)			AUC (mg.h/L)		
				Pred	Obs	FE	Pred	Obs	FE
Karjagin et al. [18]	Plasma	Short Infusion	1.3	47.19	81.43	1.72	152.72	216.94	1.42
Roberts et al. [17]	Plasma	Short Infusion	2.1	97.60	88.78	0.91	52.25	85.92	1.64
	Subcutis		2.13	48.75	32.42	0.67	73.75	38.59	0.52
	Plasma	Continuous infusion	1.12	38.21	37.39	0.98	465.61	450.88	0.96
	Subcutis		1.75	13.51	8.90	0.66	334.88	249.48	0.74

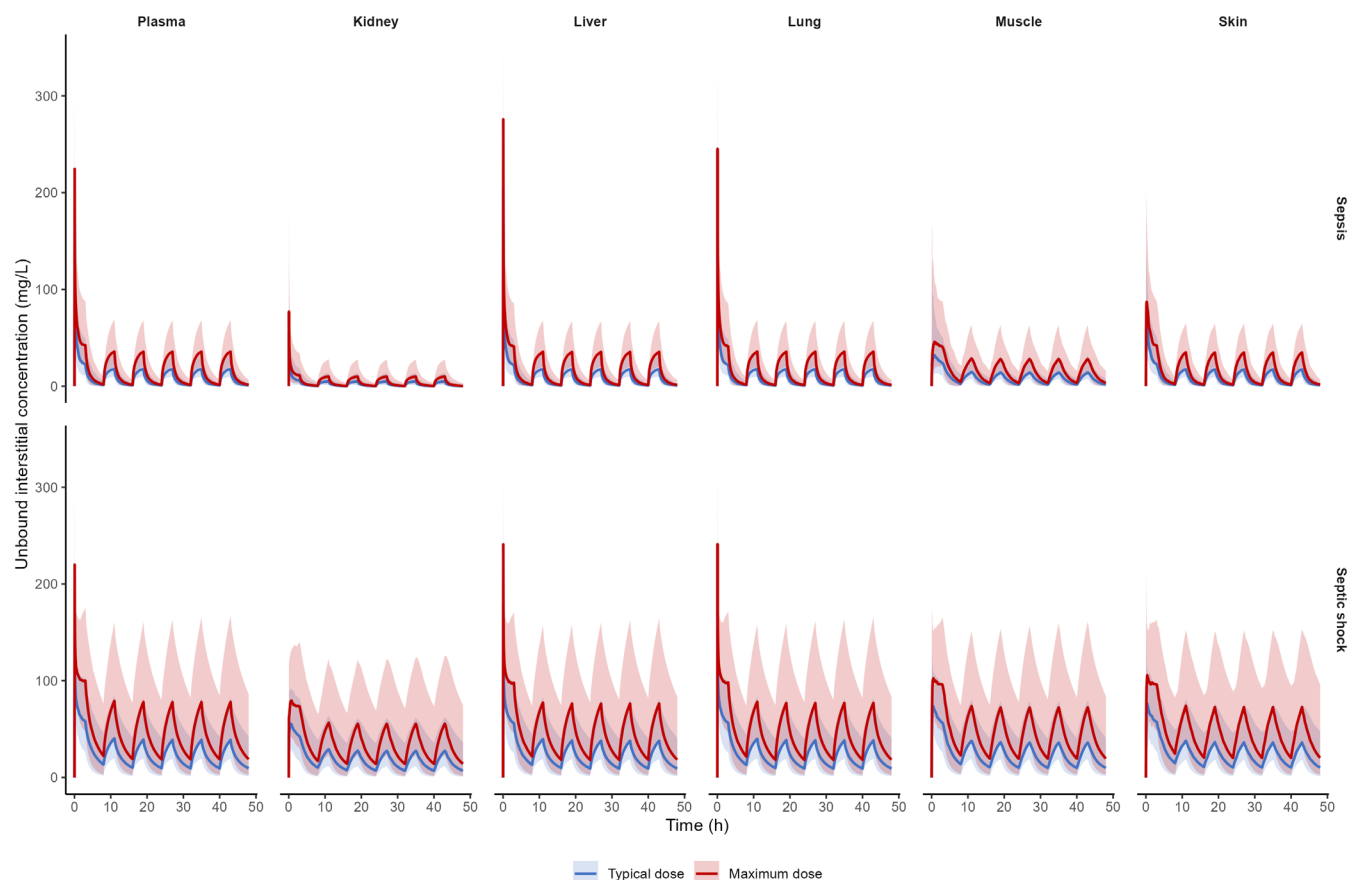


FIGURE 2 | Predicted unbound plasma and interstitial meropenem concentration–time profiles across tissues under different clinical dosing strategies. Solid lines represent the median predicted concentration and shaded areas represent the 90% prediction interval (5th–95th percentile) across the simulated population ($n = 100$). Typical dose (loading 1.5 g + maintenance 1 g, 3 h infusion, q8h) and maximum dose (loading 2 g + maintenance 2 g, 3 h infusion, q8h).

maximum dose recommended by ILAS [23] increased both $C_{\max}$ and AUC. Predicted plasma $C_{\max}$ values in sepsis ranged approximately from 165 to 218 mg/L under typical dosing and increased to values approaching 157–293 mg/L with maximum dosing, with corresponding increases in AUC. CL estimates were higher in sepsis, consistent with augmented renal elimination. Predicted values for $C_{\max}$, AUC, CL, and Vd are summarized in Table S3.

CL was reduced in septic shock (0.12 L/h/kg) compared with sepsis (0.3 L/h/kg), resulting in prolonged systemic exposure. Comparison of plasma and tissue concentration–time profiles demonstrated a clear dissociation between systemic exposure and interstitial tissue exposure across all scenarios. In sepsis, the maximum ILAS-recommended dose resulted in higher and more sustained unbound interstitial concentrations across all evaluated tissues compared with typical dosing.

PD simulations based on the $E_{\max}$ model (Figure 3) indicated that the standard MPN dosing regimen produced antibacterial effects close to the maximal achievable response in most tissues and disease states. Increases in drug exposure beyond those achieved with standard dosing resulted in only minimal additional antibacterial effect in the median scenario. However, quantification of therapeutic failure, defined as bacterial load exceeding $4 \log_{10}$ CFU/mL at 12 h, revealed that outcomes were not uniform across the simulated population. In septic

shock, therapeutic failure was 0% across all tissues and doses, consistent with the sustained drug exposure associated with reduced renal clearance. In sepsis, therapeutic failure under typical dosing was most pronounced in the kidney (33.3%), with residual failure observed in lung, liver, and skin (6.5%, 6.5%, and 1.4%, respectively). Dose escalation to the maximum ILAS-recommended regimen eliminated therapeutic failure in all tissues except the kidney, where the failure rate remained 33.3% regardless of dose, indicating that higher plasma exposure does not translate into improved pharmacodynamic outcomes in this compartment under conditions of ARC.

4 | Discussion

This study applied a mechanistic PBPK framework able to characterize the tissue distribution and pharmacodynamic response (PD) of MPN in a sepsis or septic shock setting, considering interstitial exposure, providing a physiological explanation for the variability in systemic clearance observed in critically ill patients. The simulations demonstrated heterogeneous tissue exposure of the drug, which did not have a relevant impact on the antibacterial effect.

A key finding was the distinct behavior of MPN clearance (CL) in the settings of sepsis and septic shock. In sepsis, CL was

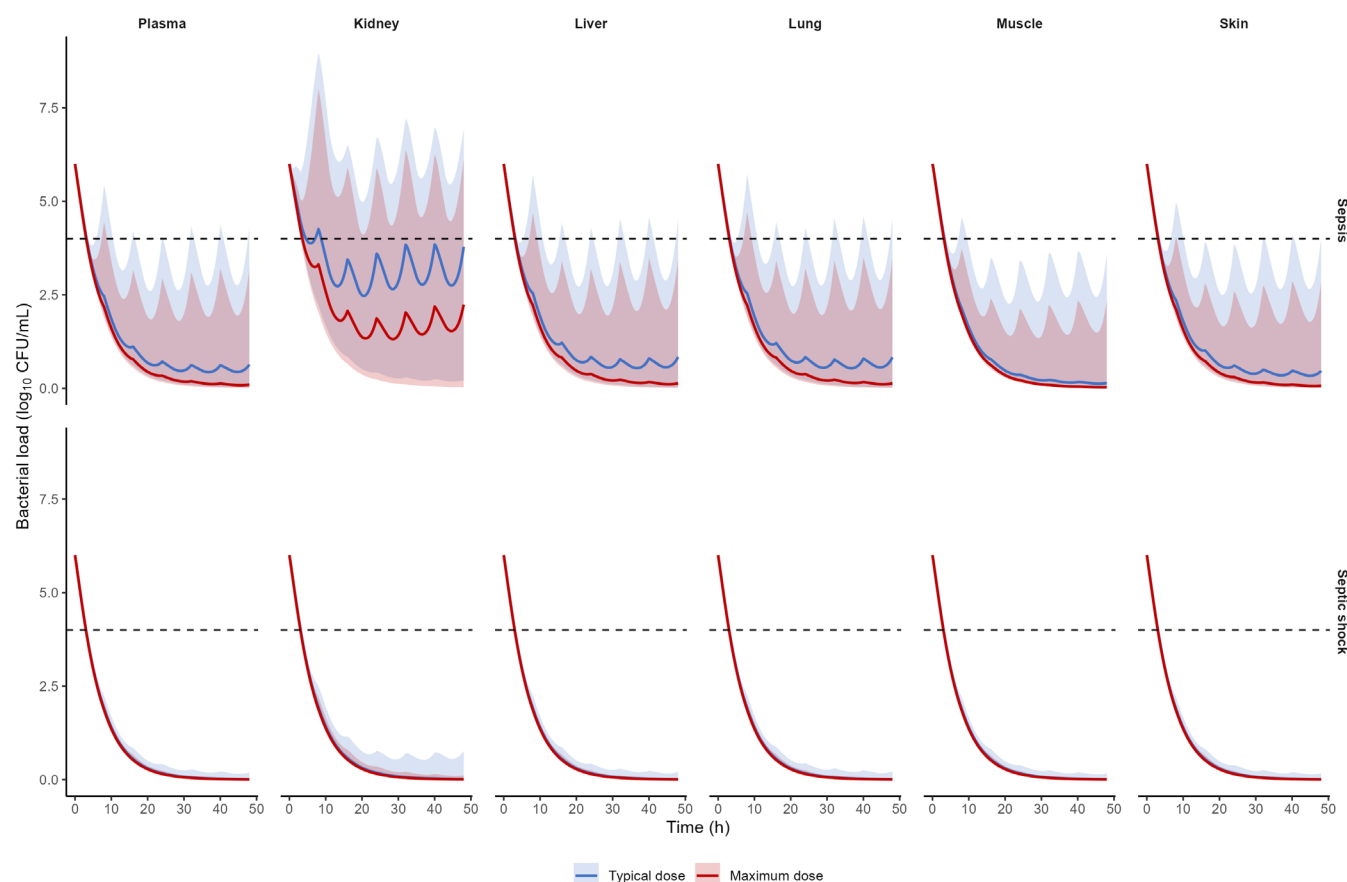


FIGURE 3 | Predicted pharmacodynamic response over time in plasma and interstitial tissues following different clinical dosing strategies. Solid lines represent the median predicted bacterial load and shaded areas represent the 90% prediction interval (5th–95th percentile) across the simulated population ($n = 100$). Typical dose (loading 1.5 g + maintenance 1 g, 3 h infusion, q8h) and maximum dose (loading 2 g + maintenance 2 g, 3 h infusion, q8h).

increased—an event associated with the hyperdynamic phase characterized by elevated cardiac output and renal perfusion, leading to ARC and potentially resulting in subtherapeutic exposure with standard dosing regimens [25, 26]. Conversely, septic shock was associated with reduced CL, reflecting the transition to sustained hypotension, renal hypoperfusion, and acute kidney injury (AKI)—conditions in which both the glomerular filtration rate (GFR) and tubular secretory capacity are impaired [27, 28].

Recently, Bahnasawy et al. [29] developed a whole-body PBPK model demonstrating good agreement with observed plasma and subcutaneous tissue concentrations across sepsis stages, providing important insights into the contribution of body composition changes, interstitial fluid expansion, and renal function to altered MPN disposition [29]. The present study builds upon these findings by incorporating tissue-resolved exposure and an integrated PD component, complementing that framework through explicit representation of individual elimination pathways to support mechanistic interpretation across the sepsis spectrum.

MPN renal elimination is not solely determined by glomerular filtration but also involves active tubular secretion mediated by OAT3, operating as a mechanistically distinct and parallel pathway [8]. Dong et al. demonstrated that MPN is a substrate

of OAT1/3, with the two pathways operating in parallel rather than redundantly [8], and Hunt et al. showed that incorporating OAT3-specific activity profiles substantially improved predictive performance beyond GFR-based scaling alone [20]. The present SA further confirmed that GFR fraction and OAT3 k_{cat} exert quantitatively distinct and tissue-specific effects on MPN exposure, with GFR fraction dominating systemic CL and OAT3 k_{cat} contributing primarily at the renal level, sensitivity profiles inconsistent with compensatory fitting and reflective of mechanistically separable elimination pathways. Furthermore, OAT3 activity is independently downregulated by proinflammatory cytokines during sepsis, independently of changes in GFR [30], a disease-state-dependent modulation of tubular secretion that cannot be captured by aggregate CL terms and represents a mechanistic determinant of interstitial drug exposure accessible only through explicit transporter modeling.

The SA of P_{end} revealed that AUC_{inf} was largely insensitive to permeability variation across tissues and disease states, supporting the robustness of the pharmacodynamic conclusions to uncertainty in this parameter. The opposing direction of C_{max} sensitivity between disease states and tissues reflects the distinct hemodynamic contexts governing plasma-tissue exchange in each disease state [31, 32]. The kidney represented a mechanistic exception, exhibiting positive AUC_{inf} sensitivity

to P_{end} -Kidney in sepsis: increased permeability-driven substrate delivery to the renal interstitium occurs in the context of inflammation-induced OAT3 downregulation [9, 30], which may reduce tubular secretory capacity and limit drug removal from the interstitial compartment, resulting in net interstitial accumulation. This behavior underscores the mechanistic interaction between P_{end} and transporter-mediated elimination in organs with active secretory function, and highlights why explicit transporter representation is necessary to predict tissue-level drug exposure in this compartment.

The present findings reinforce the concept that plasma-based PK assessment is insufficient to characterize antimicrobial efficacy in critically ill patients. As highlighted by Gonzalez et al. drug efficacy for anti-infective agents should be related to unbound concentrations at the site of action, since disease- and drug-related factors can lead to substantial differential tissue distribution, rendering plasma concentrations an unreliable surrogate for target site exposure [33]. This is particularly important in sepsis, where tissue-specific alterations in perfusion and permeability further decouple systemic from interstitial drug availability, and where MIC-based breakpoints derived from plasma PK may not adequately reflect drug availability at the infection site.

Dose escalation using maximum ILAS-recommended regimens increased plasma and tissue exposure but did not uniformly improve pharmacodynamic outcomes across all tissues or disease states. Quantification of therapeutic failure revealed that in septic shock, failure was 0% across all tissues and doses, reflecting sustained drug exposure under reduced renal clearance. In sepsis, therapeutic failure under typical dosing was most pronounced in the kidney (33.3%), with residual failure in lung, liver, and skin (6.5%, 6.5%, and 1.4%, respectively). Dose escalation eliminated therapeutic failure in all tissues except the kidney, where the failure rate remained 33.3% regardless of dose. This persistence of renal therapeutic failure despite increased plasma exposure reflects the ARC characteristic of the hyperdynamic sepsis phase, which accelerates systemic drug elimination and limits the ability of dose escalation to sustain adequate renal interstitial concentrations over time. For most tissues, the PD analysis based on the E_{max} model indicated that antibacterial effect was already close to the plateau of the concentration–effect relationship under standard dosing, such that increases in exposure translated into minimal additional effect [24, 34, 35]. In the PK simulation, standard dosing appeared insufficient only for resistant Gram-negative pathogens with MIC > 32 mg/L, with peak C_{ss} of 34 mg/L.

Although standard dosing achieved near-maximal antibacterial effect in the median scenario for most tissues, outcomes were not uniform across the simulated population. Interindividual variability was substantial in sepsis, and patients with significant ARC remained at risk of therapeutic failure even under maximum dosing, especially in the kidney. In these patients, prolonged or continuous infusion strategies may be advantageous by stabilizing interstitial drug exposure over time, while dose individualization guided by therapeutic drug monitoring would be expected to confer greater pharmacodynamic benefit than uniform dose escalation [17, 36].

Moreover, given MPN short half-life and rapid attainment of steady state, the PK benefit of a loading dose appears limited, as early increases in plasma concentrations did not translate into sustained improvements in tissue exposure. Safety considerations further support a cautious approach to universal dose escalation, since excessive systemic exposure in the context of reduced clearance has been associated with neurotoxicity, particularly in patients with renal dysfunction [37].

However, this study is based on some assumptions that should be highlighted. The disease-specific OAT3 k_{cat} values represent optimized fitting parameters rather than independently validated transporter activity estimates. Although consistent with experimental evidence of cytokine-mediated transporter downregulation in sepsis [30], quantitative measurement of OAT3 activity in critically ill patients remains unavailable. Similarly, although cumulative urinary recovery information from healthy volunteer studies was used to validate the mechanistic partitioning between glomerular filtration and active tubular secretion, equivalent data were not available in the clinical studies used for model scaling in sepsis and septic shock. Independent validation of renal pathway partitioning against urinary excretion endpoints in critically ill patients therefore remains an important gap, and future studies incorporating urine collection protocols alongside plasma and tissue sampling would substantially strengthen the mechanistic interpretability of transporter-informed PBPK models in this population. In addition, the PBPK/PD framework does not explicitly account for host immune response, and therefore cannot directly predict clinical outcomes such as infection resolution or survival, a limitation partially mitigated by the integration of a mechanistic PD model enabling direct evaluation of the concentration–effect relationship. Finally, while the present work extends PK/PD assessment to tissue-based exposures, PD targets are still largely defined using plasma concentrations and in vitro MIC values, and validated thresholds for interstitial tissue concentrations have yet to be established.

In conclusion, the integrated PBPK/PD approach demonstrates that MPN CL and tissue exposure change dynamically across sepsis severity, with increased clearance in sepsis driven by augmented renal clearance and reduced clearance in septic shock associated with AKI and impaired transporter-mediated secretion. Despite pronounced tissue-specific and interindividual variability, PD simulations indicate that standard dosing achieves near-maximal antibacterial effect for most tissues, but patients with significant ARC remain at risk of therapeutic failure, underscoring the limitations of plasma-only assessment and supporting individualized, mechanism-informed optimization rather than routine dose escalation for antimicrobial therapy in critically ill patients.

Author Contributions

Laura Ben Olivo wrote the manuscript. Bibiana Verlindo de Araújo designed the research. Laura Ben Olivo, Jessica Luisa Silva de Lemos, and Vinicius Jardim Rodrigues performed the research. Laura Ben Olivo analyzed the data.

Acknowledgments

The authors thank Georg Hempel (Universität Münster) for kindly providing the septic population files.

Funding

This research was funded by the National Council for Scientific and Technological Development (CNPq) #423260/2021-9. Bibiana Verlindo de Araujo is a CNPq/Brazil research fellow (process #315521/2023-6). Laura Ben Olivo received a scholarship from the Coordination for the Improvement of Higher Education Personnel (CAPES/Brazil).

Conflicts of Interest

The authors declare no conflicts of interest.

References

1. S. Dhaese, S. Van Vooren, J. Boelens, and J. De Waele, "Therapeutic Drug Monitoring of β -Lactam Antibiotics in the ICU," *Expert Review of Anti-Infective Therapy* 18, no. 11 (2020): 1155–1164, <https://doi.org/10.1080/14787210.2020.1788387>.
2. R. Pai Mangalore, A. Ashok, S. J. Lee, et al., "Beta-Lactam Antibiotic Therapeutic Drug Monitoring in Critically Ill Patients: A Systematic Review and Meta-Analysis," *Clinical Infectious Diseases* 75, no. 10 (2022): 1848–1860, <https://doi.org/10.1093/cid/ciac506>.
3. R. Pai Mangalore, T. N. Peel, A. A. Udy, and A. Y. Peleg, "The Clinical Application of Beta-Lactam Antibiotic Therapeutic Drug Monitoring in the Critical Care Setting," *Journal of Antimicrobial Chemotherapy* 78, no. 10 (2023): 2395–2405, <https://doi.org/10.1093/jac/dkad223>.
4. A. Benítez-Cano, S. Luque, L. Sorlí, et al., "Intrapulmonary Concentrations of Meropenem Administered by Continuous Infusion in Critically Ill Patients With Nosocomial Pneumonia: A Randomized Pharmacokinetic Trial," *Critical Care* 24, no. 1 (2020): 55, <https://doi.org/10.1186/s13054-020-2763-4>.
5. R. Rohani, P. R. Yarnold, M. H. Scheetz, et al., "Individual Meropenem Epithelial Lining Fluid and Plasma PK/PD Target Attainment," *Antimicrobial Agents and Chemotherapy* 67, no. 12 (2023): e0072723, <https://doi.org/10.1128/aac.00727-23>.
6. J. M. Varghese, P. Jarrett, S. C. Wallis, et al., "Are Interstitial Fluid Concentrations of Meropenem Equivalent to Plasma Concentrations in Critically Ill Patients Receiving Continuous Renal Replacement Therapy?," *Journal of Antimicrobial Chemotherapy* 70, no. 2 (2015): 528–533, <https://doi.org/10.1093/jac/dku413>.
7. M. Sistanizad, R. Hassanpour, and E. Pourheidari, "Are Antibiotics Appropriately Dosed in Critically Ill Patients With Augmented Renal Clearance? A Narrative Review," *International Journal of Clinical Practice* 2022 (2022): 1867674, <https://doi.org/10.1155/2022/1867674>.
8. J. Dong, Y. Liu, L. Li, Y. Ding, J. Qian, and Z. Jiao, "Interactions Between Meropenem and Renal Drug Transporters," *Current Drug Metabolism* 23, no. 5 (2022): 423–431, <https://doi.org/10.2174/1389200223666220428081109>.
9. N. Karimian Pour and M. Piquette-Miller, "Dysregulation of Renal Transporters in a Rodent Model of Viral Infection," *International Immunopharmacology* 80 (2020): 106135, <https://doi.org/10.1016/j.intimp.2019.106135>.
10. N. Isoherranen, "Physiologically Based Pharmacokinetic Modeling of Small Molecules: How Much Progress Have We Made?," *Drug Metabolism and Disposition* 53, no. 1 (2025): 100013, <https://doi.org/10.1124/dmd.123.000960>.
11. R. P. Bax, W. Bastain, A. Featherstone, D. M. Wilkinson, M. Hutchison, and S. J. Haworth, "The Pharmacokinetics of Meropenem in Volunteers," *Journal of Antimicrobial Chemotherapy* 24, no. Suppl A (1989): 311–320, https://doi.org/10.1093/jac/24.suppl_a.311.
12. B. A. Christensson, I. Nilsson-Ehle, M. Hutchison, S. J. Haworth, B. Oqvist, and S. R. Norrby, "Pharmacokinetics of Meropenem in Subjects With Various Degrees of Renal Impairment," *Antimicrobial Agents and Chemotherapy* 36, no. 7 (1992): 1532–1537, <https://doi.org/10.1128/AAC.36.7.1532>.
13. W. A. Krueger, J. Bulitta, M. Kinzig-Schippers, et al., "Evaluation by Monte Carlo Simulation of the Pharmacokinetics of Two Doses of Meropenem Administered Intermittently or as a Continuous Infusion in Healthy Volunteers," *Antimicrobial Agents and Chemotherapy* 49, no. 5 (2005): 1881–1889, <https://doi.org/10.1128/AAC.49.5.1881-1889.2005>.
14. B. Ljungberg and I. Nilsson-Ehle, "Pharmacokinetics of Meropenem and Its Metabolite in Young and Elderly Healthy Men," *Antimicrobial Agents and Chemotherapy* 36, no. 7 (1992): 1437–1440, <https://doi.org/10.1128/AAC.36.7.1437>.
15. A. Leroy, J. P. Fillastre, F. Borsa-Lebas, I. Etienne, and G. Humbert, "Pharmacokinetics of Meropenem (ICI 194,660) and Its Metabolite (ICI 213,689) in Healthy Subjects and in Patients With Renal Impairment," *Antimicrobial Agents and Chemotherapy* 36, no. 12 (1992): 2794–2798, <https://doi.org/10.1128/AAC.36.12.2794>.
16. I. Nilsson-Ehle, M. Hutchison, S. J. Haworth, and S. R. Norrby, "Pharmacokinetics of Meropenem Compared to Imipenem-Cilastatin in Young, Healthy Males," *European Journal of Clinical Microbiology & Infectious Diseases* 10, no. 2 (1991): 85–88, <https://doi.org/10.1007/BF01964413>.
17. J. A. Roberts, C. M. Kirkpatrick, M. S. Roberts, T. A. Robertson, A. J. Dalley, and J. Lipman, "Meropenem Dosing in Critically Ill Patients With Sepsis and Without Renal Dysfunction: Intermittent Bolus Versus Continuous Administration? Monte Carlo Dosing Simulations and Subcutaneous Tissue Distribution," *Journal of Antimicrobial Chemotherapy* 64, no. 1 (2009): 142–150, <https://doi.org/10.1093/jac/dkp139>.
18. J. Karjagin, S. Lefeuvre, K. Oselin, et al., "Pharmacokinetics of Meropenem Determined by Microdialysis in the Peritoneal Fluid of Patients With Severe Peritonitis Associated With Septic Shock," *Clinical Pharmacology and Therapeutics* 83, no. 3 (2008): 452–459, <https://doi.org/10.1038/sj.clpt.6100312>.
19. Automeris, "WebPlotDigitizer," (2023), <https://apps.automeris.io/wpd/>.
20. J. P. Hunt, S. Dubinsky, M. K. AM, et al., "Maximum Likelihood Estimation of Renal Transporter Ontogeny Profiles for Pediatric PBPK Modeling," *CPT: Pharmacometrics & Systems Pharmacology* 13, no. 4 (2024): 576–588, <https://doi.org/10.1002/psp4.13102>.
21. C. Radke, D. Horn, C. Lanckohr, et al., "Development of a Physiologically Based Pharmacokinetic Modelling Approach to Predict the Pharmacokinetics of Vancomycin in Critically Ill Septic Patients," *Clinical Pharmacokinetics* 56, no. 7 (2017): 759–779, <https://doi.org/10.1007/s40262-016-0475-3>.
22. H. P. Gavin, "The Levenberg-Marquardt Method for Nonlinear Least Squares Curve-Fitting Problems," (2016), <https://people.duke.edu/~hpgavin/ce281/lm.pdf>.
23. Latin American Sepsis Institute, *Practical Guide to Antimicrobial Therapy in Sepsis*, 2nd ed. (Latin American Sepsis Institute, 2022).
24. R. Saporta, E. I. Nielsen, J. U. Hansen, E. Liepinsh, I. K. Minichmayr, and L. E. Friberg, "PK/PD Modelling and Simulation of Longitudinal Meropenem in Vivo Effects Against Escherichia Coli and Klebsiella pneumoniae Strains With High MICs," *International Journal of Antimicrobial Agents* 64, no. 6 (2024): 107389, <https://doi.org/10.1016/j.ijantimicag.2024.107389>.
25. A. A. Udy, J. A. Roberts, R. J. Boots, D. L. Paterson, and J. Lipman, "Augmented Renal Clearance: Implications for Antibacterial Dosing in the Critically Ill," *Clinical Pharmacokinetics* 49, no. 1 (2010): 1–16, <https://doi.org/10.2165/11318140-000000000-00000>.

26. A. A. Udy, J. M. Varghese, M. Altukroni, et al., "Subtherapeutic Initial β -Lactam Concentrations in Select Critically Ill Patients: Association Between Augmented Renal Clearance and Low Trough Drug Concentrations," *Chest* 142, no. 1 (2012): 30–39, <https://doi.org/10.1378/chest.11-1671>.
27. F. Pea and P. Viale, "Bench-To-Bedside Review: Appropriate Antibiotic Therapy in Severe Sepsis and Septic Shock-Does the Dose Matter?," *Critical Care* 13, no. 3 (2009): 214, <https://doi.org/10.1186/cc7774>.
28. E. A. J. Hoste, J. A. Kellum, N. M. Selby, et al., "Global Epidemiology and Outcomes of Acute Kidney Injury," *Nature Reviews. Nephrology* 14, no. 10 (2018): 607–625, <https://doi.org/10.1038/s41581-018-0052-0>.
29. S. M. Bahnasawy, N. J. Parrott, M. Gijzen, I. Spriet, L. E. Friberg, and E. I. Nielsen, "Physiologically-Based Pharmacokinetic Modelling in Sepsis: A Tool to Elucidate How Pathophysiology Affects Meropenem Pharmacokinetics," *International Journal of Antimicrobial Agents* 64, no. 6 (2024): 107352, <https://doi.org/10.1016/j.ijantimicag.2024.107352>.
30. Y. P. Tsang, T. Hao, Q. Mao, E. J. Kelly, and J. D. Unadkat, "Dysregulation of the mRNA Expression of Human Renal Drug Transporters by Proinflammatory Cytokines in Primary Human Proximal Tubular Epithelial Cells," *Pharmaceutics* 16, no. 2 (2024): 285, <https://doi.org/10.3390/pharmaceutics16020285>.
31. C. Joukhadar, M. Frossard, B. X. Mayer, et al., "Impaired Target Site Penetration of Beta-Lactams May Account for Therapeutic Failure in Patients With Septic Shock," *Critical Care Medicine* 29, no. 2 (2001): 385–391, <https://doi.org/10.1097/00003246-200102000-00030>.
32. P. De Paep, F. M. Belpaire, and W. A. Buylaert, "Pharmacokinetic and Pharmacodynamic Considerations When Treating Patients With Sepsis and Septic Shock," *Clinical Pharmacokinetics* 41, no. 14 (2002): 1135–1151, <https://doi.org/10.2165/00003088-200241140-00002>.
33. D. Gonzalez, S. Schmidt, and H. Derendorf, "Importance of Relating Efficacy Measures to Unbound Drug Concentrations for Anti-Infective Agents," *Clinical Microbiology Reviews* 26, no. 2 (2013): 274–288.
34. W. A. Craig, "Pharmacokinetic/Pharmacodynamic Parameters: Rationale for Antibacterial Dosing of Mice and Men," *Clinical Infectious Diseases* 26, no. 1 (1998): 1–12, <https://doi.org/10.1086/516284>.
35. V. H. Tam, P. S. McKinnon, R. L. Akins, M. J. Rybak, and G. L. Drusano, "Pharmacodynamics of Cefepime in Patients With Gram-Negative Infections," *Journal of Antimicrobial Chemotherapy* 50, no. 3 (2002): 425–428, <https://doi.org/10.1093/jac/dkf130>.
36. A. A. Udy, J. A. Roberts, and J. Lipman, "Clinical Implications of Antibiotic Pharmacokinetic Principles in the Critically Ill," *Intensive Care Medicine* 39, no. 12 (2013): 2070–2082, <https://doi.org/10.1007/s00134-013-3088-4>.
37. G. An, C. B. Creech, N. Wu, et al., "Evaluation of Empirical Dosing Regimens for Meropenem in Intensive Care Unit Patients Using Population Pharmacokinetic Modeling and Target Attainment Analysis," *Antimicrobial Agents and Chemotherapy* 67, no. 1 (2023): e0131222, <https://doi.org/10.1128/aac.01312-22>.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Pharmacodynamic parameters used in the simulations. Values from Saporta et al. (2024). **Figure S1:** Goodness-of-fit plot for the PBPK model in healthy volunteers. **Figure S2:** Goodness-of-fit plot for the PBPK model in patients with sepsis (Roberts et al. 2009b) and septic shock (Karjagin et al. 2008). **Figure S3:** Sensitivity analysis of glomerular filtration rate (GFR fraction) and OAT3 catalytic turnover rate (kcat) on meropenem plasma and interstitial tissue exposure in sepsis and septic shock. Each parameter was varied independently over a range of $\pm 50\%$ relative to its optimized value, with all other parameters held constant. **Figure S4:** Sensitivity analysis of endothelial permeability (Pendo) across simulated tissues on meropenem plasma and interstitial tissue exposure in sepsis and septic

shock. Each tissue-specific Pendo parameter was varied independently over a range of $\pm 50\%$ relative to the optimized base value, with all other parameters held constant. **Table S2:** Summary of study designs and patient characteristics of the studies in healthy volunteers included in the validation of the PBPK model. **Table S3:** Pharmacokinetics parameters predicted by the PBPK model for clinically recommended empirical doses in septic patients.